Optimal Experiments for Partial Causal Effect Identification

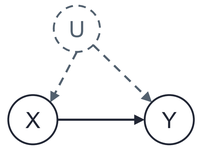
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Choose experiments by how much they guarantee to tighten causal bounds.

Tighter bounds within a budget

Observational data may only bound a causal effect. Experiments are costly. Which experiments guarantee the greatest tightening of bounds within a budget?

Guaranteed tightening of bounds



θ = P(Y_1 = 1, Y_0 = 0)

Probability that treatment is necessary and sufficient for the outcome (PNS).
Experiment a: learn P(Y | do(X = 1)).

Observational bounds: 0 ≤ θ ≤ 0.6



Guaranteed tightening of θ's bounds from experiment a: 0.6 − 0.5 = 0.1

Epistemic potency

$$W(a) = \max_{\text{feasible } p} [U(a, p) - L(a, p)]$$

a is a set of experiments; p is their joint outcome.
U and L are the resulting upper and lower bounds.

$$pot(a) = W - W(a)$$

W is the observational bound width.

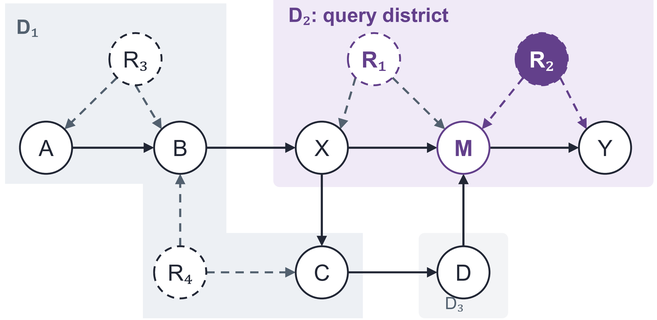
$$\max_{a \subseteq A} pot(a) \quad \text{subject to} \quad c(a) \leq B$$

A: candidate experiments; c(a): cost; B: budget.
Subset selection is NP-hard; potency uses polynomial programs [1].

What can the graph rule out?

Query θ = P(Y = 1 | do(M = 1))

In this graph, a district is a connected component using only dashed edges, ignoring their direction.



Relevant response-types have a directed path to Y that does not go through M. Here, only R_2 does.

Relevant response-types: $R_\theta = \{R_2\}$

The district containing R_2 also contains R_1, so the district hull includes both response-types.

District hull: $R^*_\theta = \{R_1, R_2\}$

1 Path interception

If every directed path from the district hull to an experiment's outcomes goes through its interventions, that experiment has zero potency.

P(D | do(C)): all paths to D go through C → pot = 0.



Linear time per candidate: O(|V| + |E|).

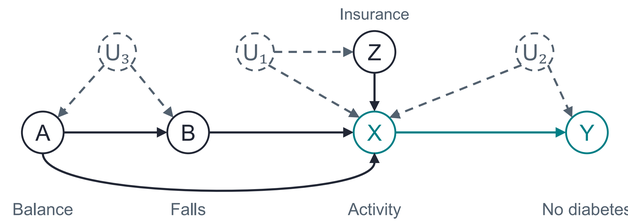
This certifies singletons only: a discarded candidate may still help when combined with other experiments.

2 ID-inertness for subset selection

If an experiment is identifiable from observational data, it has zero potency. Adding it to any combination of experiments leaves that combination's potency unchanged. It can therefore be removed before subset selection.

Physical activity & diabetes

NHANES 2017–2018: 5,833 adults.
Target: ATE of activity on not having diabetes.



Observational ATE: [−0.445, 0.419]

Candidate experiments and potency

Experiment	What is learned	Cost	Potency
a_1 P(X | do(B))	Activity after intervening on falls	1	**0**
a_2 P(B | do(A))	Falls after intervening on balance	1	**0**
a_3 P(X,Y | do(Z=1))	Activity and outcome with insurance	1	**0**
a_4 P(Y_0=1, Y_1=0)	Counterfactual harm probability	1	**0.44**
a_5 P(Y | do(X=0))	Outcome without activity	2	**0.36**
a_6 P(Y | do(X=1))	Outcome with activity	2	**0.50**

For singleton selection, remove a_1 (identifiable) and a_2 (path interception): both have zero potency.

Best choices by budget B

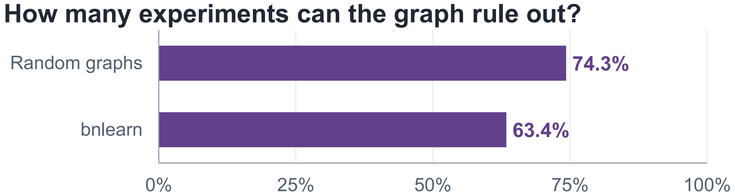
B = 1 → a_4
Potency 0.44

B = 2 → a_6
Potency 0.50

B = 3 → a_4 + a_5
Potency 0.78

Potency is not additive: (0.78 ≠ 0.44 + 0.36)

How many experiments can the graph rule out?



Singleton candidates pruned (mean): 11 bnlearn networks, 1,100 runs; random graphs, 1,596 runs. No solver calls in the pruning step.
Subset selection: ID-inertness removes 60.0% of bnlearn candidates and 61.4% on random graphs (mean).

≥6.95× less solver time
Singleton selection; 26 graphs, timeout-limited benchmark.

[1] Duarte et al. Automated causal inference in discrete settings. JASA, 2024.
[2] Shpitser & Pearl. Identification of joint interventional distributions. AAAI, 2006.
Tian & Pearl. Probabilities of causation: Bounds and identification.
Annals of Mathematics and Artificial Intelligence 28:287–313, 2000.

Limitations: known graph, finite discrete variables and exact P(V).
Bounding programs are costly; early stopping may yield non-sharp bounds.
Pruning is incomplete; interception is not safe for subset removal.
Other informativeness criteria can select different experiments.

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Paper PDF

GitHub repo